

Software Requirements Specification for Keystone: Capstone Project Management System

Team 13, Keystone

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Revision History

Date	Version	Notes
Date 1	1.0	Notes
Date 2	1.1	Notes

1 Introduction

1.1 Purpose

Software Engineering is intended to solve ...

1.2 Scope

TODO

1.3 Stakeholders

- Customer
- End Users
- Project Sponsor
- Development Team
- Administrators

1.4 Definitions and Acronyms

Term	Definition
TODO	TODO

Table 1: Definitions and Acronyms

1.5 Characteristics of Intended Reader

2 System Overview

This section provides general information about the system. It identifies the problem, the vision, and the interfaces between the system and its environment. This section also describes user characteristics and lists the system constraints.

2.1 Problem Statement

TODO

2.2 Product Vision

TODO

2.3 System Context

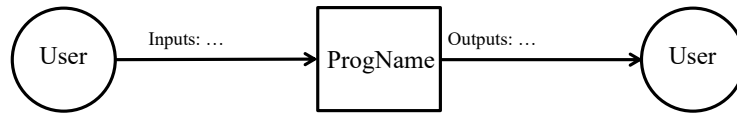


Figure 1: System Context

- User Responsibilities:

-

- Software Engineering Responsibilities:

- Detect data type mismatch, such as a string of characters instead of a floating point number

-

2.4 User Characteristics

2.5 System Constraints

3 Stakeholder Requirements

SR1 TODO

SR2 TODO

SR3 TODO

4 User Personas

4.1 Persona 1

- Name:
- Description:
- Goals:
- Frustrations:
- Technical Ability:

4.2 Persona 2

- Name:
- Description:
- Goals:
- Frustrations:
- Technical Ability:

5 Use Cases

5.1 UC1: Example Use Case

Primary Actor

TODO

Preconditions

- TODO

Main Success Scenario

1. TODO
2. TODO
3. TODO

Alternative Flows

- TODO

Postconditions

- TODO

6 Functional Requirements

FR1: The system shall ...

FR2: The system shall ...

FR3: The system shall ...

7 Data Requirements

7.1 Major Data Entities

- User
- Song
- Session
- Report

7.2 Data Dictionary

Attribute	Type	Description
user_id	UUID	Unique identifier
email	String	User email address

Table 2: Data Dictionary

7.3 Privacy and Data Retention

TODO

8 Non-Functional Requirements

NFR1: **Accuracy**

NFR2: **Usability**

NFR3: **Maintainability**

NFR4: **Portability**

- Other NFRs that might be discussed include verifiability, understandability, reusability, performance, reliability and availability, security, and accessibility. You may have to add other NFRs.

9 AI and Machine Learning Requirements

9.1 Model Inputs

TODO

9.2 Model Outputs

TODO

9.3 Evaluation Metrics

TODO

9.4 Known Limitations

TODO

10 Requirements Traceability Matrices

Stakeholder Req	Use Case	Functional Req
SR1	UC1	FR1

Table 3: Traceability Between Stakeholder Requirements, Use Cases, and Functional Requirements

Functional Req	Nonfunctional Req
FR1	NFR1

Table 4: Traceability Between Functional and Nonfunctional Requirements

11 Rationale

12 Likely Changes

LC1:

13 Unlikely Changes

LC2:

14 Appendices

14.1 Appendix A: Additional Materials

TODO

14.2 Appendix B: Development Plan

14.3 Appendix C: Values of Auxiliary Constants

TODO

14.4 Appendix D: Generative AI Disclosure

14.5 Appendix E: Reflection

The information in this section will be used to evaluate the team members on the graduate attribute of Lifelong Learning.

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?
4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.
5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be

related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.

6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?